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Assignment 2 Question 2

Due: 10/10/2025

2a)

FCFS

Process ID	Completion Time	Turnaround Time	Waiting Time
J1	$0+10=10$	$10-0=10$	$10-10=0$
J2	$10+8=18$	$18-2=16$	$16-8=8$
J3	$18+3=21$	$21-3=18$	$18-3=15$
J4	$21+4=25$	$25-10=15$	$15-4=11$
J5	$25+1=26$	$26-12=14$	$14-1=13$
J6	$26+4=30$	$30-15=15$	$15-4=11$
Average	21.667	14.667	9.667

2b)

SJF

Process ID	Completion Time	Turnaround Time	Waiting Time	Notes
J1	$0+10=10$	$10-0=10$	$10-10=0$	J2-J4 arrive at or before J1 CT
J2	$22+8=30$	$30-2=28$	$28-8=20$	
J3	$10+3=13$	$13-3=10$	$10-3=7$	J5 has arrived
J4	$14+4=18$	$18-10=8$	$8-4=4$	J6 has arrived
J5	$13+1=14$	$14-12=2$	$2-1=1$	
J6	$18+4=22$	$22-15=7$	$7-4=3$	
Average	17.833	9.333	5.833	

2b)

Preemptive SJF

Process ID	Completion Time	Turnaround Time	Waiting Time
J1	$0+13=13$	$13-0=13$	$13-10=3$
J2	$22+8=30$	$30-2=28$	$28-8=20$
J3	$3+3=6$	$6-3=3$	$3-3=0$
J4	$14+4=18$	$18-10=8$	$8-4=4$
J5	$13+1=14$	$14-12=2$	$2-1=1$
J6	$18+4=22$	$22-15=7$	$7-4=3$
Average	17.166	10.166	5.166

J1 runs until $T=3$ and J3 starts, this is because at $T=2$ the remaining time on J1 and J2 is the same so J1 continues to run, J1 has $T=7$ time left.

J3 runs from $T=3$ to $T=6$ and J1 resumes.

J1 runs from $T=6$ to $T=13$ since at $T=10$ J1 has $T=3$ time remaining and J4 has $T=4$ time left, at $T=12$ both J1 and J5 have the same remaining time. So J1 continues to run

J5 runs from $T=13$ to $T=14$.

J4 runs from $T=14$ to $T=18$.

J6 runs from $T=18$ to $T=22$.

J2 runs from $T=22$ to $T=28$.

2c)

Process ID	Completion Time	Turnaround Time	Wait Time
J1	24	$24-0=24$	$(9-3)+(18-10)=14$
J2	29	$29-2=27$	$(3-2)+(14-6)+(25-15)=17$
J3	9	$9-3=6$	$6-3=3$
J4	25	$25-10=15$	$24-13=11$
J5	14	$14-12=2$	$13-12=1$
J6	30	$30-15=15$	$29-18=11$
Average	21.833	14.833	9.5

J1 starts running at T=0 in Q1 and runs until T=3 when it is demoted to Q2, J1 RT=7.

J2 and J3 have arrived in this time slice and are both in Q1.

Q1:J2->J3

J2 runs from T=3 to T=6 and is demoted to Q2, J2 RT=5.

Q2:J1->J2

J3 runs from T=6 to T=9, J3 has completed its job.

J1 runs from T=9 to T=10 when J4 enters and starts running, J1 RT=6 and J1 is placed last in Q2.

Q2:J2->J1

J4 runs from T=10 to T=13 and is demoted to Q2, J4 RT=1. J5 has arrived during this time and is placed in Q1.

Q2:J2->J1->J4

J5 runs from T=13 to T=14, J5 has completed its job.

J2 runs from T=14 to T=15 when J6 enters and starts running, J2 RT=4 and J2 is placed last in Q2.

Q2:J1->J4->J2

J6 runs from T=15 to T=18 and is demoted to Q2, J6 RT=1.

Q2:J1->J4->J2->J6

J1 runs from T=18 to T=24 and has completed its job.

J4 runs from T=24 to T=25 and has completed its job.

J2 runs from T=25 to T=29 and has completed its job.

J6 runs from T=29 to T=30 and has completed its job.

